

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Cache Organization

Lecture No.- 9

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Recap of Previous Lecture



Topic

Array Access With Cache



Topics to be Covered



Topic

Multilevel Cache

Topic

Cache Inclusion Policy

#Q. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

How many data cache misses will occur in total?



#Q. Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

- A** line 4 to line 11
- B** line 4 to line 12
- C** line 0 to line 7
- D** line 0 to line 8



#Q. Consider a direct mapped write back data cache of size 2KB with block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array $A[30][25]$ with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 4 times. And between the accesses, there is no any data cache changes happen. Hit ratio of cache for the array access is?

#Q. A CPU has a 32KB direct mapped cache with 128 byte-block size. Suppose A is two-dimensional array of size 512×512 with elements that occupy 8-bytes each. Consider the following two C code segments, P1 and P2.

P1:

```
for (i = 0 ; i < 512 ; i++)  
{  
    for (j = 0 ; j < 512 ; j++)  
    {  
        x += A[i][j] ;  
    }  
}
```

P2:

```
for (i = 0 ; i < 512 ; i++)  
{  
    for (j = 0 ; j < 512 ; j++)  
    {  
        x += A[j][i] ;  
    }  
}
```

#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P_1 be M_1 and that for P_2 be M_2 .

The value of M_1 is :



#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P_1 be M_1 and that for P_2 be M_2 .

The value of M_2 is :



#Q. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

- A** Every access to S is a hit.
- B** Once P is brought to the cache it is never evicted.
- C** At the end of the execution only R and S reside in the cache.
- D** Every access to R evicts Q from the cache.



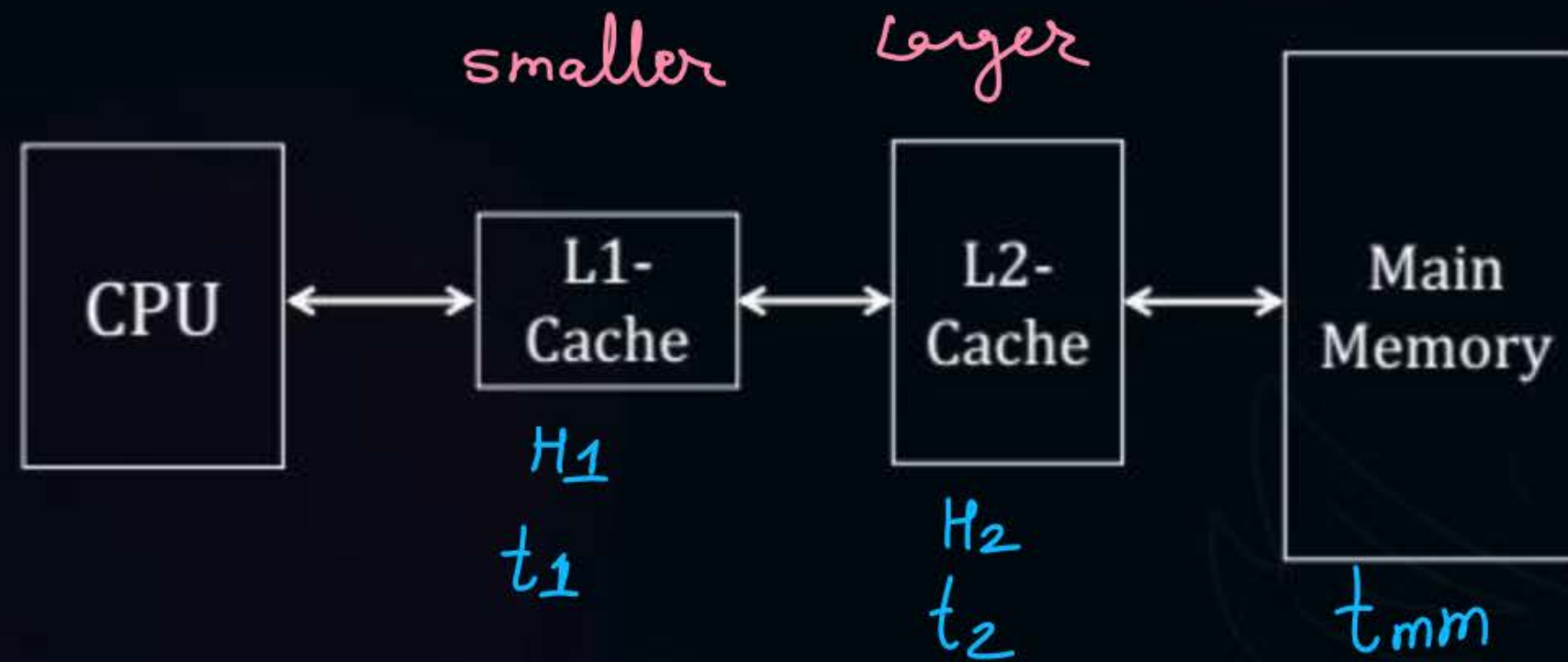
Topic : Goal of Using Cache

1. Minimize Access Time $\rightarrow$ smaller cache
 2. Maximize Hit Rate $\rightarrow$ larger cache
 3. Minimize Miss Penalty
- } $\Rightarrow$ multilevel cache





Topic : Multilevel Cache





Topic : Average Access Time Multilevel Cache

Simultaneous:-

$$t_{avg} = H_1 * t_1 + (1 - H_1) \left[H_2 * t_2 + (1 - H_2) t_{mm} \right]$$

or

$$= H_1 t_1 + (1 - H_1) H_2 t_2 + (1 - H_1)(1 - H_2) t_{mm}$$

(Default)
Hierarchical:-

$$t_{avg} = H_1 t_1 + (1-H_1) \left[H_2 (t_1 + t_2) + (1-H_2) (t_1 + t_2 + t_{mm}) \right]$$

or

$$= H_1 t_1 + (1-H_1) H_2 (t_1 + t_2) + (1-H_1)(1-H_2) (t_1 + t_2 + t_{mm})$$

or

$$= t_1 + (1-H_1) \left[t_2 + (1-H_2) t_{mm} \right]$$

#Q. Consider a 3-level memory hierarchy with L1 cache, L2 cache and a main memory. The hit ratios of L1 is 90% and of L2 is 95%. The access times of L1, L2 and main memory are 15ns, 60ns and 350ns respectively. The average memory access time is _____ns?

$$T_{avg} = 15 + 0.1 \left[60 + 0.05 * 350 \right] \\ = 22.75 \text{ ns}$$

if in this mem. system we use only L1 and mm.

$$t_{avg} = 15 + 0.1 * 350 = 50 \text{ ns}$$

if in this mem. system we use only L2 and mm.

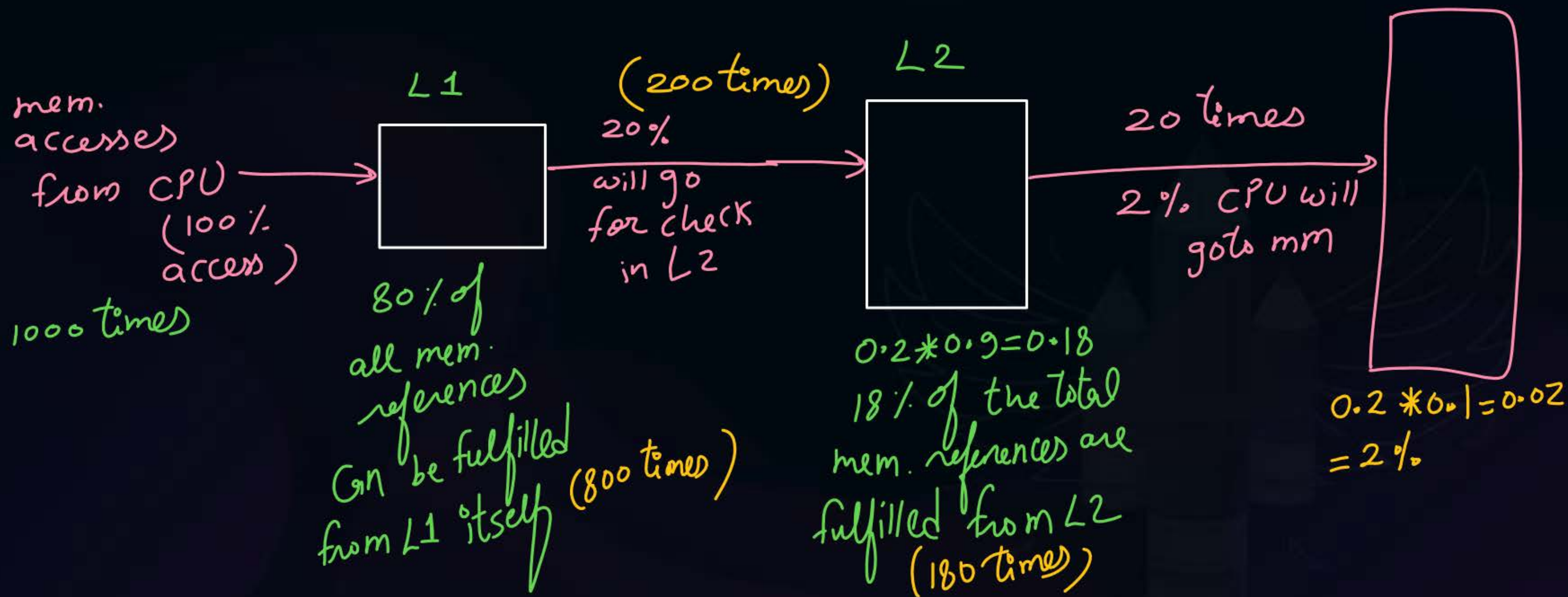
$$t_{avg} = 60 + 0.05 * 350 = 77.5 \text{ ns}$$



Topic : Probability of Access

Assume $H_1 = 80\%$

$H_2 = 90\%$



Probability of mem.
access

% of mem references are fulfilled from L1 = H_1

_____ || _____ $L_2 = (1 - H_1) H_2$

_____ || _____ $mm = (1 - H_1)(1 - H_2)$

#Q. Consider a 3-level memory hierarchy with L1 cache, L2 cache and a main memory. The probability of access of L1 is 95%, of L2 is 4.5% and of main memory is 0.5%. The access times of L1, L2 and main memory are 10ns, 50ns and 400ns respectively. The average memory access time is 14.5 ns?

$$= 0.95 * 10 + 0.045 * (10 + 50) + 0.005 * (10 + 50 + 400)$$

$$= 14.5 \text{ ns}$$



Topic : Dual Cache



$$\begin{aligned}
 t_{avg \text{ inst}^n} &= H_{1i} t_{1i} + (1 - H_{1i}) \left[H_2 * (t_{1i} + t_2) + (1 - H_2) (t_{1i} + t_2 + t_{mm}) \right] \\
 &\quad \text{or} \\
 &= t_{1i} + (1 - H_{1i}) \left[t_2 + (1 - H_2) t_{mm} \right]
 \end{aligned}$$

$$\begin{aligned}
 t_{avg \text{ data}} &= H_{1d} t_{1d} + (1 - H_{1d}) \left[H_2 * (t_{1d} + t_2) + (1 - H_2) (t_{1d} + t_2 + t_{mm}) \right] \\
 &\quad \text{or} \\
 &= t_{1d} + (1 - H_{1d}) \left[t_2 + (1 - H_2) t_{mm} \right]
 \end{aligned}$$

$$t_{avg} = \% \text{ of data access} * t_{avg \text{ data}} + \% \text{ of inst}^n \text{ access} * t_{avg \text{ inst}^n}$$

#Q. The multilevel memory hierarchy is given.



The hit ratio of $L1_i$, $L1_d$, $L2$ and main memory are 0.8, 0.9, 0.95 and 1.0 respectively. The access times of respective memories are 10ns, 10ns, 50ns and 500ns. Among total memory references 60% of them are for data.

1. Average memory access time for only instructions access 25 ns
2. Average memory access time for only data access 17.5 ns
3. Average memory access time 20.5 ns

$$1. \text{tavg}^{\text{inst}} = 10 + 0.2 \left[50 + 0.05 * 500 \right] = 25 \text{ ns}$$

$$2. \text{tavg}^{\text{data}} = 10 + 0.1 \left[50 + 0.05 * 500 \right] = 17.5 \text{ ns}$$

$$\begin{aligned} 3. \text{tavg} &= 0.6 * 17.5 \text{ ns} + 0.4 * 25 \\ &= 20.5 \text{ ns} \end{aligned}$$

#Q. The read access times and the hit ratios for different caches in a memory hierarchy are as given below:

Cache	Read access time (in nanoseconds)	Hit Ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the write-back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____?

$$T_{avg}^{inst^h} = 2 + 0.2 \left[8 + 0.1 * 90 \right] = 5.4 \text{ ns}$$

$$T_{avg}^{data} = 2 + 0.1 \left[8 + 0.1 * 90 \right] = 3.7 \text{ ns}$$

$$\begin{aligned} T_{avg} &= 0.6 * 5.4 + 0.4 * 3.7 \\ &= 4.72 \text{ ns} \end{aligned}$$

#Q. Consider a program execution which has 36% instructions for load and store. The CPI without memory stalls is 2. The program experiences 2% miss for instruction cache and 4% miss for data cache. The cache miss penalty is 200 cycles.

1. CPI with memory stalls $\Rightarrow 8.88$
2. Performance gain (speed up) of perfect cache as compared to cache with stalls $\frac{8.88}{2} = 4.44$

$$CPI = 2 + \text{extra cycles per inst}^n \text{ for inst}^n \text{ access} + \text{extra cycles per inst}^n \text{ for data access}$$

$$= 2 + (1.0 * 0.02 * 200) + (0.36 * 0.04 * 200)$$

$$= 8.88$$



Topic : Cache Inclusion Policy



↓
mem blocks which are in $L1$ should be
present in $L2$ or not?





Topic : Inclusion Policy

(inclusive policy)

L_2 must be inclusive of L_1 .

(all mm blocks which are present in L_1 , must be present in L_2 also).

$$(\text{content of } L_1) \cap (\text{content of } L_2) \Rightarrow \text{content of } L_1$$

Value inclusion policy:-

block of L_1 will always be present in L_2 with same values.



Topic : Inclusion Policy



cpu wants to read value x from mem.

1. Hit in L1 *cpu reads x from L1*

2. Miss in L1 & Hit in L2 *cpu reads x from L2*

cpu copies the missed block from L2 to L1.

If any block is replaced from L1 then there is no any role of L2.

3. Miss in L1 & Miss in L2 *cpu reads x from mm*

cpu copies missed block from mm to L2 then from L2 to L1.

If any block is replaced from L2 then L2 sends a back validation signal to L1 to make this replaced block invalid in L1.



Topic : Exclusion Policy

all mm blocks which are present in $L1$, must not be present in $L2$.

$$(\text{Content of } L1) \cap (\text{Content of } L2) \Rightarrow \emptyset$$

→ $L2$ cache is called as victim cache ⇒

↳ $L2$ keeps blocks which are replaced from $L1$.



Topic : Exclusion Policy

CPU wants to read x from mem.

1. Hit in L1 CPU reads x from L1

2. Miss in L1 & Hit in L2 CPU reads x from L2

The missed block removed from L2 and copied to L1.

If any block is replaced from L1 then move it to L2.

3. Miss in L1 & Miss in L2 CPU reads x from mm

The missed block is copied from mm to L1.

If any block is replaced from L1 then move it L2.



2 mins Summary



Topic

Multilevel Cache

Topic

Cache Inclusion Policy





Happy Learning

THANK - YOU

